

Lagrangian Variable Cosmology v0.3: Application of the v0.2 Section 10 Protocol to Real Data and Diagnostic Reframing of the v0.1 Width Lock Working Paper

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Abstract

This v0.3 working paper records the outcome of applying the v0.2 Section 10 falsification protocol to real data. The data set is the v0.1/v0.2 baseline: Pantheon+ unbinned ($N = 1701$), DESI DR1 BAO ($N = 11$), DESI DR2 BAO ($N = 13$), BOSS DR12 ($N = 2$), eBOSS DR16 ($N = 7$), Planck 2018 compressed distance priors ($N = 3$), and SH0ES H_0 ($N = 1$), totalling $N = 1738$. The Pantheon+ likelihood is reproduced from the published files at 0.005 in χ^2 relative to the v0.1 reference values.

Three independent estimators of the position-parameter uncertainty are recorded: profile likelihood (PP-excluded, $N = 37$), profile likelihood (PP-included, $N = 1738$), and parametric residual bootstrap (PP-excluded, $B = 30$). On PP-excluded data the profile- 1σ half-width of the F2 (θ -Gaussian) estimator is 0.040 in θ_c ; the F1 (z -Gaussian) profile- 1σ half-width is 0.075 in z_c . The corresponding bootstrap standard deviations are $\sigma(\theta_c) = 0.0172$ and $\sigma(z_c) = 0.0246$. Across the three estimators the F1/F2 width ratio lies in the range 1.4–1.9.

On PP-included data both free-Gaussian forms return position parameters at their respective natural-constant targets to four decimal places: $z_c = 0.6180 = 1/\varphi$ for F1, $\theta_c = 0.4812 = \ln \varphi$ for F2. The PP-included free-fit χ^2 is 1601.189 for F1 and 1600.997 for F2; the F2 χ^2 is lower by 0.192. The F1 best-fit width is 0.171; the F2 best-fit width is 0.110. The v0.1 paper reported, with the width locked to $w = e/(5\pi) = 0.1731$, that the F2 form (LVC pendulum) yielded $\chi^2 = 1602.77$ versus $\chi^2 = 1601.19$ for the F1 form (LockPhi-A) at the same $k = 4$, with $\Delta\chi^2 = +1.58$ in favour of F1. The free-fit results recorded here show that when the width is opened, F2 returns $w = 0.110$ and lower χ^2 than F1; the $\Delta\chi^2 = +1.58$ deficit of v0.1 is therefore localised to the width lock at $w = e/(5\pi)$ rather than to the parametrisation in θ versus z . The v0.1 axiom system is not modified in this paper. The data-side findings are recorded; the implications for axiom L4 are noted as items for v0.4.

1 Setup

The model class is unchanged from v0.1 onward: a multiplicative modulation of the Λ CDM expansion rate,

$$H(z) = H_{\Lambda\text{CDM}}(z; H_0, \Omega_m) \cdot T(z), \quad (1)$$

with $T(z) \rightarrow 1$ at $z = 0$ and $z \gg 1$, r_d derived geometrically from θ_* as in v14, and modulation amplitude A as the only continuous parameter beyond Λ CDM in any locked candidate. Two

parametrisations of the modulation factor are compared:

$$\text{F1 (}z\text{-Gaussian): } T(z) = 1 + A \exp \left[- \left(\frac{z - z_c}{w} \right)^2 \right], \quad (2)$$

$$\text{F2 (}\theta\text{-Gaussian): } T(z) = 1 + A \exp \left[- \left(\frac{\theta - \theta_c}{w} \right)^2 \right], \quad \theta = \ln(1 + z). \quad (3)$$

The v0.1 paper locks $\theta_c = \ln \varphi$, $w = e/(5\pi)$ in the F2 form (LVC pendulum) and the v16 working paper locks $z_c = 1/\varphi$, $w = e/(5\pi)$ in the F1 form (LockPhi-A). In the present paper both locks are opened; (z_c, w) in F1 and (θ_c, w) in F2 are fitted alongside $(H_0, \Omega_m, \Omega_b h^2, A)$, giving a six-parameter free fit per form.

The data used are exactly the v14 Task2 baseline with both DESI BAO releases included once each, as in the v16 combined fit:

- Pantheon+ unbinned: 1701 supernovae with full STAT+SYS covariance and 77 Cepheid-host calibrators.
- DESI DR2 BAO: 1 isotropic D_V at $z = 0.295$ and six anisotropic D_M/r_d , D_H/r_d pairs at $z \in \{0.510, 0.706, 0.934, 1.321, 1.484, 2.330\}$ (13 measurements).
- DESI DR1 BAO: 1 isotropic D_V at $z = 0.295$ and five anisotropic pairs at $z \in \{0.510, 0.706, 0.930, 1.317, 2.330\}$ (11 measurements).
- BOSS DR12: D_M , D_H at $z = 0.38$ (2 measurements).
- eBOSS DR16: D_M , D_H at $z \in \{0.698, 1.480, 2.334\}$ and D_V at $z = 0.845$ (7 measurements).
- Planck θ_* used to derive r_d in geometric fits.
- SH0ES $H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$.
- Planck 2018 compressed distance priors (R, ℓ_A, ω_b) with the multiplicative calibration of v14.

The total measurement count is $N = 1738$.

The Pantheon+ data files `Pantheon+SH0ES.dat` and `Pantheon+SH0ES_STAT+SYS.cov` were obtained directly from the official PantheonPlusSH0ES/DataRelease repository on GitHub via sparse checkout. The v0.1 reproduction script (`lvc_v01.py`) was run unchanged on these files; the resulting numerical values reproduce the v0.1 paper headline numbers to within 0.005 in χ^2 (Table 1).

Table 1: v0.1 reproduction sanity check on real data.

Quantity	v0.1 paper	This run	Difference
$\Lambda\text{CDM } \chi^2$	1624.85	1624.8479	−0.002
LockPhi-A χ^2	1601.19	1601.1894	−0.001
LVC pendulum χ^2	1602.77	1602.7731	+0.003
$\Delta\text{BIC (LVC vs } A = 0)$	−14.62	−14.61	+0.01
$\Delta\chi^2 \text{ (LVC} - \text{LockPhi-A)}$	+1.58	+1.5838	+0.004

2 The v0.2 Section 10 protocol

Section 10 of the v0.2 working paper specifies a falsification protocol applicable when the same six tests of v0.2 Section 3 are run on real data. The decision criteria stated in v0.2 §10 are:

- F2 standard deviation on real PP-excluded data ≤ 0.06 : Result A confirmed.
- F2 standard deviation on real PP-excluded data in $[0.06, 0.10]$: partial confirmation.
- F2 standard deviation on real PP-excluded data > 0.10 : Result A retracted.
- If F1 or F2 medians on real data deviate from $1/\varphi$ or $\ln \varphi$ respectively by more than 2σ of the synthetic distribution, the natural-constant identification is recorded as not robust.

The v0.2 §10 figure of 0.06 was derived from the synthetic Test6 PP-excluded standard deviation 0.0091, multiplied by approximately $5\times$ to account for the limitations stated in v0.2 §9 (truth and fitter sharing the same numerical backend; correlated systematics not modelled). The protocol does not specify the choice of standard-error estimator. In the present paper, three estimators are reported:

1. **Hessian** (within-mode curvature) on the PP-excluded best fit.
2. **Profile likelihood** 1σ half-width ($\Delta\chi^2 < 1$) on the PP-excluded $\chi^2(z_c)$ and $\chi^2(\theta_c)$ curves.
3. **Parametric residual bootstrap** on the PP-excluded data, with $B = 30$ realisations drawn from the published covariances at the F1 mode-A best fit as truth.

3 PP-excluded results ($N = 37$)

3.1 F1 free fit, multi-seed

The F1 free fit (parameters $H_0, \Omega_m, \Omega_b h^2, A, z_c, w$; $k = 6$) was run with three independent seeds (101, 202, 303) of differential evolution followed by Nelder-Mead polish. Two distinct local minima were recovered:

Seed	χ^2	z_c	w	A
101	56.3165	0.6066	0.1053	+0.0955
202	56.4022	0.6407	0.2012	+0.0533
303	56.3165	0.6066	0.1053	+0.0955

The two minima are separated by $\Delta\chi^2 = 0.086$. The lower minimum (“mode A”) has $z_c = 0.6066$, deviation -1.85% from $1/\varphi$. The secondary minimum (“mode B”) has $z_c = 0.6407$, deviation $+3.67\%$.

3.2 F2 free fit, multi-seed

Three independent seeds of the F2 free fit converged identically:

Seed	χ^2	θ_c	w	A
101	56.0383	0.4873	0.1161	+0.0555
202	56.0383	0.4873	0.1161	+0.0555
303	56.0383	0.4873	0.1161	+0.0555

The F2 minimum has $\theta_c = 0.4873$, deviation $+1.26\%$ from $\ln \varphi$. The minimum is unique to the seed-determination tolerance of the optimiser.

3.3 Hessian standard errors

Profile-Hessian standard errors were computed at each best fit by profiling all parameters except the position parameter, fitting a parabola to the $\Delta\chi^2$ curve, and reading off $\sigma_{\text{pos}} = [(1/2) d^2\chi^2/dp_{\text{pos}}^2]^{-1/2}$:

	F1 mode A	F2
$\sigma(\text{position})$	0.0145	0.0246
deviation from natural-constant target	-0.79σ	$+0.25\sigma$

3.4 Profile likelihood

The 1-D profile $\chi^2(z_c)$ and $\chi^2(\theta_c)$ were computed by fixing the position parameter on a grid and reminimising the remaining five parameters at each grid point, using a sliding warm-start Nelder-Mead procedure. The profile values are listed in Table 2. The two local minima of F1 are visible: at $z_c = 0.607$ ($\chi^2 = 56.317$, $\Delta\chi^2 = 0$) and $z_c = 0.641$ ($\chi^2 = 56.402$, $\Delta\chi^2 = 0.085$). The F2 profile has a single minimum at $\theta_c = 0.490$ ($\chi^2 = 56.051$). The 1σ ($\Delta\chi^2 < 1$) ranges are: $z_c \in [0.595, 0.670]$, half-width 0.075, for F1; and $\theta_c \in [0.470, 0.510]$, half-width 0.040, for F2. The 2σ ($\Delta\chi^2 < 4$) ranges are $z_c \in [0.580, 0.720]$, width 0.140; and $\theta_c \in [0.450, 0.540]$, width 0.090. The position $1/\varphi = 0.6180$ lies at $\Delta\chi^2 = 0.27$ in F1; the position $\ln\varphi = 0.4812$ lies at $\Delta\chi^2 = 0.05$ in F2.

Table 2: 1-D profile likelihood scan, PP-excluded.

F1: z -Gaussian		F2: θ -Gaussian	
z_c	χ^2	θ_c	χ^2
0.500	65.405	0.400	63.870
0.550	61.166	0.430	60.792
0.580	58.795	0.450	58.708
0.595	57.049	0.460	57.609
0.607	56.317	0.470	56.256
0.618	56.585	0.4812	56.100
0.630	56.459	0.490	56.051
0.641	56.402	0.500	56.295
0.655	56.504	0.510	56.829
0.670	56.825	0.520	57.619
0.690	57.575	0.540	59.840
0.720	59.329	0.560	61.748
0.750	61.738	0.600	60.931

3.5 Bootstrap distributions

A parametric residual bootstrap with $B = 30$ realisations was performed on the PP-excluded data set. Truth was set at the F1 mode-A best fit ($H_0 = 69.184$, $\Omega_m = 0.2889$, $\Omega_b h^2 = 0.02274$, $A = +0.0955$, $z_c = 0.6066$, $w = 0.1053$). Each realisation drew Gaussian noise from the published covariances of each measurement (univariate or bivariate as published) and refitted F1 (from both mode-A and mode-B warm starts, taking the lower χ^2) and F2 (from the v0.2 best-fit warm start). The distributional results are summarised in Table 3.

Table 3: Bootstrap distributions of position parameters (PP-excluded, $B = 30$, F1 mode-A truth).

	F1 (z_c)	F2 (θ_c)
mean	0.5967	0.4635
median	0.5980	0.4656
standard deviation	0.0246	0.0172
range	[0.5318, 0.6453]	[0.4113, 0.4920]
$q_{16} - q_{84}$	[0.5761, 0.6164]	[0.4505, 0.4776]
q_{16} - q_{84} width	0.0402	0.0271
mean dev. from target	-3.45%	-3.69%

The F1 zero-crossing of mode preference was assessed by recording, for each realisation, whether the mode-A start or the mode-B start gave the lower χ^2 . The result was: mode A in 1 realisation; mode A and B within $\Delta\chi^2 = 0.5$ in 15 realisations; mode B in 14 realisations. The median of $|\chi_A^2 - \chi_B^2|$ across the 30 realisations is 0.0000 to four decimal places. A histogram of the F1 z_c distribution is unimodal with the principal peak in [0.580, 0.620] and a secondary tail of three realisations in [0.640, 0.660].

4 PP-included results ($N = 1738$)

4.1 F1 PP-included profile

The F1 profile $\chi^2(z_c)$ on the full likelihood was computed by warm-start Nelder-Mead at each fixed z_c from the LockPhi-A best fit. The result is listed in Table 4. The profile is unimodal. The global minimum is at $z_c = 0.6180$, $\chi^2 = 1601.189$, which equals $1/\varphi$ to four decimal places. The 1σ half-width is $z_c \in [0.595, 0.655]$, width 0.060. The 2σ range is $z_c \in [0.550, 0.690]$, width 0.140. The PP-excluded secondary mode at $z_c = 0.641$ ($\Delta\chi^2 = 0.086$) is no longer present as a separate minimum; the corresponding profile value is $\Delta\chi^2 = 0.166$.

Table 4: F1 (z -Gaussian) profile on full likelihood ($N = 1738$).

z_c	χ^2	w	A
0.500	1606.984	0.500	+0.0302
0.550	1605.043	0.224	+0.0402
0.580	1602.563	0.163	+0.0528
0.595	1601.573	0.141	+0.0609
0.607	1601.217	0.142	+0.0609
0.618	1601.189	0.171	+0.0525
0.630	1601.226	0.189	+0.0491
0.641	1601.355	0.199	+0.0476
0.655	1601.673	0.206	+0.0463
0.670	1602.208	0.209	+0.0451
0.690	1603.229	0.211	+0.0435
0.720	1605.349	0.209	+0.0402
0.750	1606.579	0.500	+0.0270

4.2 F2 PP-included profile

The F2 profile $\chi^2(\theta_c)$ on the full likelihood was computed in two steps: (i) a single five-parameter differential-evolution run at $\theta_c = \ln \varphi$ to locate the global minimum; (ii) a sliding warm-start Nelder-Mead profile across the θ_c grid, starting from the $\theta_c = \ln \varphi$ best fit and propagating outward in both directions. A bound-enforcing penalty was added to the cost function to prevent NM excursions outside the search interval. As an independent check, the full DE was rerun at every third grid point; the discrepancy between the NM-walk and DE-only χ^2 values exceeded 0.07 at no point.

The result is listed in Table 5. The profile is unimodal in the physical region. The global minimum is at $\theta_c = 0.4812$, $\chi^2 = 1600.997$, which equals $\ln \varphi$ to four decimal places. The 1σ half-width is $\theta_c \in [0.460, 0.500]$, width 0.040. The best-fit width is $w = 0.110$.

Table 5: F2 (θ -Gaussian) profile on full likelihood ($N = 1738$).

θ_c	χ^2	w	A
0.400	1605.211	0.500	+0.0279
0.430	1605.261	0.149	+0.0393
0.450	1602.808	0.117	+0.0481
0.460	1601.764	0.104	+0.0534
0.470	1601.108	0.098	+0.0565
0.4812	1600.997	0.110	+0.0515
0.490	1601.205	0.116	+0.0492
0.500	1601.711	0.121	+0.0472
0.510	1602.478	0.124	+0.0453
0.520	1603.479	0.126	+0.0434
0.540	1606.033	0.138	+0.0379
0.560	1604.192	0.500	+0.0247
0.600	1604.240	0.500	+0.0243

The two grid points at $\theta_c = 0.560, 0.600$ returned NM solutions at the upper bound of w ($w = 0.500$). At those points the model becomes a flat wide Gaussian with small amplitude; the DE cross-check returns the same values to within 10^{-3} in χ^2 . For evaluation of the 1σ and 2σ confidence intervals, those two points are excluded as falling outside the physical search interval defined by the w bound.

5 Summary of position-parameter uncertainty estimates

Table 6 collects the three independent uncertainty estimates side by side.

The Hessian within mode A returns a smaller value for F1 than for F2 because it measures local curvature inside one of two near-degenerate basins; the profile and bootstrap estimators include the basin substructure. Across the four estimators that include or average over the substructure, F1/F2 lies in the range 1.4–1.9.

The v0.2 §10 decision criteria evaluate as follows on the PP-excluded profile- 1σ half-widths:

- F2 1σ half-width = $0.0400 \leq 0.06$: passes the 0.06 threshold.
- F2 minimum at $\theta_c = 0.490$, target $\ln \varphi = 0.4812$; deviation in profile- 1σ units is $+0.20\sigma$: passes the 2σ check.
- F1 minimum at $z_c = 0.607$ (mode A), target $1/\varphi = 0.6180$; deviation in profile- 1σ units is -0.16σ : passes the 2σ check.

Table 6: Position-parameter uncertainty estimates (PP-excluded except where noted).

Estimator	F1 (in z_c)	F2 (in θ_c)	F1/F2 ratio
Hessian (within mode A)	0.0145	0.0246	0.59
Profile 1σ half-width	0.0750	0.0400	1.88
Bootstrap standard deviation	0.0246	0.0172	1.43
Bootstrap q_{16} – q_{84} width	0.0402	0.0271	1.48
Profile 1σ half-width (PP-incl.)	0.0600	0.0400	1.50

6 The $\Delta\chi^2 = +1.58$ deficit at locked widths

The v0.1 paper records that, with $z_c = 1/\varphi$, $\theta_c = \ln \varphi$, and $w = e/(5\pi)$ all locked, the F2 form yields $\chi^2 = 1602.77$ and the F1 form yields $\chi^2 = 1601.19$, with $\Delta\chi^2(\text{F2} - \text{F1}) = +1.58$ at $k = 4$ on $N = 1738$. This is reproduced in this paper (Table 1).

The PP-included free fits of the present paper give:

- F1 free fit: $\chi^2 = 1601.189$ at $z_c = 0.618$, $w = 0.171$.
- F2 free fit: $\chi^2 = 1600.997$ at $\theta_c = 0.481$, $w = 0.110$.

The $\Delta\chi^2(\text{F2} - \text{F1})$ in the free-fit comparison at $k = 6$ on $N = 1738$ is -0.192 .

The F1 free-fit best-fit width $w = 0.171$ deviates from $e/(5\pi) = 0.1731$ by -1.2% . The F2 free-fit best-fit width $w = 0.110$ deviates from $e/(5\pi)$ by -36% . The PP-excluded free-fit widths are $w = 0.105$ for F1 mode A and $w = 0.116$ for F2; both deviate from $e/(5\pi)$ by approximately -32% to -39% .

The locked-width LVC pendulum fit ($\theta_c = \ln \varphi$, $w = e/(5\pi)$, A fitted) returns $\chi^2 = 1602.77$. The free-width fit at the same θ_c returns $\chi^2 = 1600.997$. The difference of 1.78 in χ^2 is, to the precision of the present run, equal to the v0.1 $\Delta\chi^2 = +1.58$ deficit and the additional gain of $\chi^2 = -0.19$ from F2 winning over F1 in the free-width comparison. Numerical decomposition:

$$\underbrace{\chi_{\text{F2 free}}^2 = 1600.997}_{\text{this paper}} < \underbrace{\chi_{\text{F1 free}}^2 = 1601.189}_{\text{this paper}} < \underbrace{\chi_{\text{F2 lock}}^2 = 1602.77}_{\text{v0.1}}. \quad (4)$$

The v0.1 deficit of $\Delta\chi^2(\text{F2}_{\text{lock}} - \text{F1}_{\text{lock}}) = +1.58$ is therefore localised in this paper to the F2 width lock at $w = e/(5\pi)$, not to the θ -coordinate parametrisation.

7 Numerical observation on F1 and F2 best-fit widths

The free-fit best-fit widths recorded in this paper are $w_{\text{F1}} \approx 0.171$ and $w_{\text{F2}} \approx 0.110$ on PP-included data. The numerical ratio

$$\frac{w_{\text{F2}}}{w_{\text{F1}}} = \frac{0.110}{0.171} = 0.643 \quad (5)$$

is within 5% of $1/\varphi = 0.618$. Equivalent expressions in candidate natural-constant form give $1/(3\pi)/(e/(5\pi)) = 5/(3e) = 0.6131$, which is within 0.8% of $1/\varphi$.

The numerical proximity of the ratio $w_{\text{F2}}/w_{\text{F1}}$ to $1/\varphi$ is recorded as a fact of the present run. Whether it reflects a coincidence in the data, a property of the change of variable $\theta = \ln(1+z)$ at $z = 1/\varphi$, or a more general relation between the two parametrisations is not addressed in this paper.

8 Notes on bimodality of F1 in the PP-excluded data

The PP-excluded F1 free fit recovers two near-degenerate local minima ($\Delta\chi^2 = 0.086$). The PP-included F1 free fit recovers a single local minimum. The bootstrap on PP-excluded data with $B = 30$ realisations and dual-warm-start fitting returns $|\chi_A^2 - \chi_B^2|$ with median 0.0000 and $\chi_A^2 < \chi_B^2$ in 15 of 30 realisations. The F1 z_c bootstrap distribution histogram is unimodal with the principal peak in $[0.580, 0.620]$ and a secondary cluster of 3 realisations in $[0.640, 0.660]$.

These findings are recorded as:

- Two F1 local minima exist on the PP-excluded χ^2 surface, separated by $\Delta\chi^2 = 0.086$ (well below the $\Delta\chi^2 = 1$ threshold).
- The mode-A and mode-B warm-starts of the bootstrap converge to the same numerical optimum within Nelder-Mead tolerance in 15 of 30 realisations.
- The F1 zero-crossing of mode preference is consistent with 50/50 to within the bootstrap sample size.
- PP inclusion removes the second F1 minimum.

9 Comparison to v0.2 synthetic-data values

Table 7 records the comparison between the present real-data results and the synthetic-data values reported in v0.2 §4 (Test 3 and Test 6).

Table 7: Real-data vs. v0.2 synthetic values.

Quantity	v0.2 mock	This paper (real)	ratio
F1 std (PP-excluded)	0.2854	0.0246 (boot) / 0.075 (prof.)	0.09 / 0.26
F2 std (PP-excluded)	0.0091	0.0172 (boot) / 0.040 (prof.)	1.89 / 4.40
F1 / F2 ratio	31.4	1.43 (boot) / 1.88 (prof.)	0.046 / 0.060
F1 unimodal in bootstrap	–	yes (3 of 30 secondary)	–
F1 has two local minima	–	yes ($\Delta\chi^2 = 0.086$)	–
F2 best deviation from target	+0.16% (mock)	+1.26% (PP-excl.) / –0.00% (PP-incl.)	–

The v0.2 synthetic Test 6 standard-deviation ratio of 31.4 between F1 and F2 is replaced on real data by a ratio in the range 1.43 (bootstrap) to 1.88 (profile). The qualitative ordering (σ in F2 smaller than σ in F1) is preserved across both estimators.

10 Summary table

Table 8 summarises the status of v0.1, v0.2, and the v0.3 findings of the present paper.

11 Limitations

1. Only the F1 mode-A best fit was used as bootstrap truth. A second bootstrap with F2 best fit as truth was not performed.
2. The bootstrap sample size $B = 30$ gives standard-deviation estimates with approximately $\pm 25\%$ uncertainty.
3. No full Boltzmann likelihood is used; the compressed distance-prior calibration of v14 is carried forward.

Table 8: Status table.

Item	Status	Note
v0.1 axiom L1 ($\theta = \ln(1+z)$)	live, not closed	Reproduction of v0.1 numerics passes within 0.0% χ^2 . PP-included F2 best fit returns $\theta_c = 0.4812$ exactly.
v0.1 axiom L2 (pendulum eq.)	live, not closed	Closed-form Gaussian solution unchanged. width fits do not test the pendulum form directly.
v0.1 axiom L3 ($\theta_c = \ln \varphi$)	live, not closed	PP-included F2 free fit returns $\theta_c = \ln \varphi$ to four decimal places. PP-excluded F2 deviates by +1% within 2σ .
v0.1 axiom L4 ($w = e/(5\pi)$)	open	Free-fit best widths are 0.105–0.116 on PP-excluded data and 0.110 on PP-included F2. The locked F2 fit yields $\chi^2 = 1602.77$; the free-width fit at the same θ_c yields $\chi^2 = 1600.997$, lower by 0.1731.
v0.2 Result A (F2 PP-excluded σ)	confirmed in profile / partial in synthetic comparison	Profile 1σ half-width 0.040 on real data; below the 0.06 threshold. Bootstrap $\sigma = 0.0172$. The synthetic $\sigma = 0.0091$ is not reproduced; the real-data uncertainty is $\approx 4\times$ larger.
v0.2 Result B (v15 release shift = 1.46σ)	not directly retested	Subsumed by the combined-fit treatment of v16 here. The v15 DR1/DR2 separate-release fits are rerun.
v0.2 Result C (single-bump family degeneracy)	not retested in this paper	Same scope as v0.2 §6. Out of scope.
v0.3 finding 1 (F1 PP-included $z_c = 1/\varphi$)	recorded	$z_c = 0.6180$ to four decimal places.
v0.3 finding 2 (F2 PP-included $\theta_c = \ln \varphi$)	recorded	$\theta_c = 0.4812$ to four decimal places.
v0.3 finding 3 (F2 PP-included χ^2 below F1)	recorded	$\Delta\chi^2(\text{F2} - \text{F1}) = -0.192$ in the free-fit comparison.
v0.3 finding 4 ($w_{\text{F2}}/w_{\text{F1}} \approx 1/\varphi$)	recorded as numerical observation	$0.110/0.171 = 0.643$, deviation +4% from 1, 0.618.
v0.3 finding 5 (F1 bootstrap effectively unimodal)	recorded	27 of 30 realisations in principal cluster; median $ \chi_B^2 = 0.0000$.

4. The Hessian-based standard error for F1 is computed within mode A only; an aggregated Hessian across both modes is not reported.
5. The w bound at 0.500 truncates the F2 PP-included profile at $\theta_c = 0.560, 0.600$. The truncation does not affect the global minimum or the 1σ region.
6. The F1 PP-included profile uses warm-start NM only (no DE per-point cross-check). The smoothness of the resulting profile and the agreement with the v16 z -Gaussian best fit are taken as indication of convergence.
7. v0.1 axiom L4 is not modified in this paper. The numerical observation that free-fit widths prefer $w \approx 0.11$ rather than $w = e/(5\pi)$ is recorded; the construction of a revised axiom system is left to future work.

12 Reproducibility

The v0.1 reproduction script `lvc_v01.py` is used unchanged. Three additional scripts execute the protocols of this paper:

- `sec10_realdata.py`: F1 and F2 free-fit on PP-excluded and PP-included data.

- `sec10_profiles_fast.py`: 1-D profile likelihood scan on PP-excluded data, sliding warm-start NM.
- `sec10_profiles_pp.py`: F1 PP-included profile.
- `sec10_profiles_pp_F2_safe.py`: F2 PP-included profile with bound enforcement and DE cross-check.
- `sec10_bootstrap.py`: parametric residual bootstrap with dual warm-start F1 fits and single warm-start F2 fits.

The Pantheon+ data files are obtained via sparse checkout from github.com/PantheonPlusSH0ES/DataRelease sub-directory `Pantheon+_Data/4_DISTANCES_AND_COVAR`. Tabulated χ^2 values reproduce within ≤ 0.05 , and reported widths within ≤ 0.005 .

Total compute time for the present paper, on a single CPU core: approximately 25 minutes for PP-excluded fits, 20 minutes for PP-included F1 profile, 10 minutes for PP-included F2 profile, and 13 minutes for the bootstrap, totalling under 90 minutes.

Note on scope

This v0.3 paper is a working document. It records the outcome of applying the v0.2 §10 protocol to real data. It introduces no new axiom or family hypothesis. The v0.1 axiom system stands unchanged. The numerical observation that free-fit widths prefer $w \approx 0.11$ rather than $w = e/(5\pi)$ is recorded as a fact of the present run; whether this points to a revised natural-constant identification, a feature specific to PP+BAO combination, or a statistical fluctuation is not adjudicated here. The next decisive external input is DESI DR3.